Verifying SimCLR: A Critical Analysis of Contrastive Learning Through Systematic Ablation

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Abstract

SimCLR [1] popularized contrastive learning for visual representations, claiming that composition of data augmentations, a nonlinear projection head, and large batch sizes are critical for performance. While these findings have been widely adopted, it remains unclear how robust they are in lower resource settings and across downstream uses. We reproduce SimCLR on CIFAR-10 and STL-10, and conduct controlled ablations to test four questions: (1) how much augmentation composition matters versus simpler policies, (2) does the projection head always improve representations, or can it harm transfer when downstream tasks require preserved information? (3) Are large batch sizes necessary, or can momentum-based methods like MoCo achieve similar results with smaller batches? (4) How sensitive optimal temperature is to dataset scale and architecture. We further analyze learned representations via invariance probes and compare against supervised pretraining and select recent self-supervised baselines. Overall, our results aim to verify which SimCLR design principles are robust and which are contingent on dataset scale and compute constraints.

1 Introduction

Learning visual representations without human supervision has become a central objective in computer vision. Contrastive learning has emerged as a dominant paradigm, with SimCLR [1] demonstrating that a simple framework can achieve competitive performance without labels. SimCLR's key contributions include: (i) composition of data augmentations (crop, color distortion, blur), (ii) a nonlinear projection head that improves the representation before it, (iii) temperature scaled contrastive loss, and (iv) very large batch sizes.

SimCLR's empirical findings have been widely adopted. However, it's original experiments were conducted on ImageNet with substantial computational resources, including batch sizes up to 8192. It remains unclear whether its core design choices are equally necessary in lower-resource settings such as CIFAR-10 or STL-10, where dataset scale, image resolution, and feasible batch size differ substantially.

Subsequent work has proposed alternative formulations of self-supervised learning. For example, BYOL [2] removes negative pairs, MoCo [4] reduces dependence on large batches via a momentum encoder and queue, and theoretical analyses [9] interpret contrastive learning through alignment and uniformity objectives. While these developments refine our understanding of representation learning, they do not directly test the robustness of SimCLR's original claims under controlled, small-scale conditions.

In this work, we revisit three central design questions:

- How critical is augmentation composition on small datasets?
- When does the nonlinear projection head improve downstream performance?

- How sensitive is performance to batch size and temperature?

We reproduce SimCLR on CIFAR-10 and STL-10 and conduct systematic ablations isolating each component. By doing so, we hope to evaluate the robustness of SimCLR's design principles under constrained data and compute regimes. By clarifying which components remain essential and which are contingent on scale, we provide practical guidance for applying contrastive learning in resource-limited settings.

2 Literature Review

2.1 Contrastive Learning Foundations

Contrastive objectives for learning invariant representations date back to early metric learning formulations [3]. Modern self-supervised variants include instance discrimination [10], which treated each image as its own class, and contrastive predictive coding (CPC) [8], which learns representations by predicting future latent variables in an autoregressive setup. SimCLR [1] simplified this line of work by removing memory banks and specialized architectures, showing that strong data augmentation and in-batch negatives can be sufficient given large batch sizes.

2.2 SimCLR Design Choices and Follow-up Challenges

SimCLR identified several components as central to strong performance: augmentation composition, a projection head, temperature scaling, and large numbers of negative pairs [1]. Subsequent work has refined how these components are understood and implemented.

- **Augmentations and view design:** SimCLR emphasized that performance depends strongly on the choice of composition of augmentations [1]. Later work [6] formalized view selection, showing that mutual information between views should be neither too high nor too low. This motivates our ablations that vary augmentation strength and composition on small datasets.

- **Projection Head:** SimCLR found that a nonlinear projection head improves the quality of representations used for contrastive training, while recommending the encoder representation for downstream evaluation [1]. Related analyses suggest the projection head can affect optimization and invariance, and not only act as an information bottleneck; for example, redundancy reduction objectives such as Barlow Twins [11] implicitly encourage structure embeddings without explicit negatives, providing an alternative perspective on what makes representations transferable.

- **Temperature and embedding geometry:** Temperature scaling in the NT-Xent loss changes how sharply positives are emphasized relative to negatives [1]. Wang and Isola [9] interpret contrastive learning through alignment (pulling positives together) and uniformity (spreading embeddings on the hypersphere), providing a useful lens for analyzing how temperature interacts with batch size and dataset scale.

- **Batch Size:** SimCLR claimed large batches are crucial. MoCo [**?**] uses a momentum encoder and queue to decouple batch size from negatives, achieving strong results with small batches. This directly challenges the necessity of large batches.

2.3 Reducing Dependence on Large Batches

A key practical limitation of SimCLR is its reliance on many negatives, which often requires large batches. MoCo [4] addresses this by using a momentum encoder and a queue of stored negatives, decoupling the number of negatives from batch size. This line of work motivates our comparison of small-batch SimCLR against momentum-queue alternatives under matched encoders and augmentations.

2.4 Recent SimCLR Extensions Relevant to this Study

Recent work continues to build directly on the SimCLR framework, often by modifying the objective or representation structure while preserving the augmentation-based multi-view setup.

- **Optimal-transport regularization**: SinSim [5] extends SimCLR by adding Sinkhorn (entropy-regularized optimal transport) regularization to encourage better-structured and better-dispersed embeddings, aiming to improve geometry beyond what is induced by the contrastive loss alone. This is conceptually aligned with our analysis of embedding geometry, though our focus remains on SimCLR's original components.
- **Set-based extensions for 3D medical data**: DeepSet SimCLR [7] (Pattern Recognition Letters, 2024) proposes a SimCLR-style objective adapted to 3D medical data while retaining 2D backbones, by modeling volume structure implicitly to reduce the compute cost of full 3D networks. This supports the broader theme that SimCLR's design choices can be adapted to different data regimes and constraints, motivating us to test robustness in low-resource settings.

While SimCLR's ablations were conducted primarily at ImageNet scale, fewer works isolate its design choices under small scale datasets and constrained batch sizes. Our work targets this gap by reproducing SimCLR on CIFAR-10 and STL-10 and systematically ablates (i) augmentation composition, (ii) projection head design, and (iii) batch size and temperature sensitivity, using alignment and uniformity metrics to support analysis.

3 Methodology

We follow the SimCLR [1] framework and adapt it for CIFAR-10 and STL-10 using this code repository as a guide: `https://github.com/sthalles/SimCLR`. Our base encoder is a ResNet-18 (and ResNet-50 for select experiments), with the first convolution adjusted for CIFAR-10 resolution. The projection head is a 2-layer MLP with hidden dimension 512 and output dimension 128. Representations are L2-normalized before computing the NT-Xent loss with temperature τ.

To balance experimental thoroughness with computational efficiency, we adopt a two-stage evaluation protocol:

- **Stage I (Screening):** Reduced pretraining (100–200 epochs) with a single random seed to identify high-signal trends.
- **Stage II (Confirmatory):** Full 800-epoch training with three random seeds for selected configurations to ensure robustness.

Unless otherwise stated, ablation sweeps are conducted in Stage I on CIFAR-10. Key findings are validated in Stage II on both CIFAR-10 and STL-10.

3.1 Baseline Reproduction

We first reproduce SimCLR on CIFAR-10 and STL-10 using the default augmentation pipeline: random resize crop, color jitter, random grayscale (20%), and Gaussian blur (30%). Baseline confirmatory runs are trained for 800 epochs with batch size 512 using the LARS optimizer and cosine learning rate decay. Screening experiments use reduced training (200 epochs).

To evaluate representation quality, we freeze the encoder and train a linear classifier for 100 epochs. We report top-1 accuracy averaged over three random seeds for confirmatory runs and one seed for screening runs.

3.2 Augmentation Ablation

To isolate the contribution of augmentation composition, we evaluate:

- Full augmentation pipeline (default SimCLR).
- Crop only.
- Crop + color jitter.

- Removal of individual augmentations.

All other hyperparameters remain fixed. These experiments are conducted in the screening stage (200 epochs, one seed). The best-performing and most informative configurations are re-evaluated in the confirmatory stage.

3.3 Projection Head Variants

To examine the role of the projection head, we compare:

- No projection head.
- Linear projection head.
- Two-layer MLP (default).
- Three-layer MLP.

For each configuration, we evaluate downstream performance using both the encoder representation $\mathbf{h}$ and the projected representation $\mathbf{z}$. Additionally, we train linear probes to predict applied augmentation parameters from each representation to assess retained transformation information. Projection head variants are first evaluated in the screening stage (200 epochs), with key findings confirmed using full 800-epoch training.

3.4 Batch Size and Temperature Sensitivity

We study the interaction between batch size and temperature by sweeping:

$$\tau \in \{0.1, 0.2, 0.5, 1.0\}$$

and

$$N \in \{64, 128, 256, 512\}.$$

This grid allows us to isolate high-level trends while controlling computational cost. These experiments are conducted in the screening stage (100 epochs, single seed).

We report linear evaluation accuracy as well as alignment and uniformity metrics [9]. Confirmatory experiments are performed for representative low-batch and high-batch settings identified in screening.

3.5 Momentum-Based Alternative

To test whether large batch sizes are fundamentally necessary, we compare SimCLR with MoCo v2 [4] under matched encoders and augmentation settings. MoCo uses a momentum encoder and queue to increase the number of negatives without increasing batch size. We compare performance at small batch sizes (e.g., 64 and 128) to determine whether queue-based methods mitigate batch-size sensitivity. Initial comparisons are performed in the screening stage, with confirmatory runs conducted for selected configurations.

4 Experimental Setup

4.1 Datasets

We evaluate on CIFAR-10 and STL-10. CIFAR-10 contains 50,000 training and 10,000 test images of size 32×32 across 10 classes. STL-10 contains 100,000 unlabeled images and 13,000 labeled images of size 96×96. For self-supervised pretraining, we use all available unlabeled images. Linear evaluation is performed using the labeled training split and evaluated on the standard test set.

4.2 Training Details

All models are implemented in PyTorch, following the SimCLR framework [1] with the public implementation by [**?**] as reference.

We use:

- Optimizer: LARS for self-supervised training.
- Learning rate scaling: $\text{lr} = 0.3 \times \text{batch_size}/256$.
- Weight decay: 10^{-6}.
- Cosine decay schedule with 10-epoch warmup.
- Total pretraining epochs: 800 (unless otherwise specified).

For linear evaluation, the encoder is frozen and a single fully-connected classifier is trained for 100

epochs using SGD with momentum 0.9. No data augmentation is applied during linear evaluation beyond standard normalization.

All reported results are averaged over three random seeds.

4.3 Compute Resources

Experiments are conducted on NVIDIA A100 GPUs (40GB memory). For batch size 512, training fits within GPU memory without gradient accumulation. Smaller batch experiments (e.g., 64 and 128) are run directly. When simulating larger effective batch sizes, we use gradient accumulation.

Estimated Compute Requirements. For CIFAR-10 with ResNet-18 and batch size 512, an 800-epoch SimCLR pretraining run requires approximately:

- $\sim$100 iterations per epoch ($50{,}000/512 \approx 98$),
- $\sim$800 epochs,
- $\sim$3.5–4.0 A100 GPU-hours per run (empirically estimated).

To reduce overall compute, we adopt a two-stage protocol. Screening runs use reduced pretraining (100–200 epochs), which scales approximately linearly in cost and typically requires $\sim$0.5–1.0 GPU-hours per configuration. Confirmatory runs use full 800-epoch training with three seeds for a small number of selected configurations.

Under this protocol, our core experimental plan consists of:

- **Screening:** $\sim$20 configurations at 100–200 epochs (one seed), totaling $\sim$15–20 GPU-hours.
- **Confirmatory:** baseline reproduction plus a small number of selected re-runs at 800 epochs (three seeds), totaling $\sim$30–40 GPU-hours.

Overall, we estimate $\sim$50–80 A100 GPU-hours for the full study, allowing additional headroom for reruns and debugging.

Supercomputing Resources. To support parallel execution and reproducibility, we plan to request GPU allocation through the Pittsburgh Supercomputing Center (PSC).

At our projected scale, the total request of $\sim$50–80 A100 GPU-hours corresponds to:

$$50\text{–}80 \text{ A100 GPU-hours} \approx 2.1\text{–}3.3 \text{ GPU-days}.$$

5 Expected Results and Discussion

We evaluate whether SimCLR's core design choices remain robust under small-scale and constrained compute settings. Our expected outcomes align with the hypotheses introduced earlier.

- **Baseline Reproduction.** We expect to achieve linear evaluation accuracy above 85% on CIFAR-10 under full 800-epoch training, consistent with prior reports. Successful reproduction establishes a reliable baseline for controlled ablations.
- **Augmentation Composition.** We expect augmentation composition to remain important, but with reduced marginal gains compared to ImageNet-scale experiments. In particular, crop-only training may perform competitively after sufficient training, suggesting that augmentation sensitivity is dataset-dependent.
- **Projection Head.** We anticipate that the nonlinear projection head improves contrastive optimization stability and overall representation quality. However, linear probes may reveal that encoder representations $\mathbf{h}$ retain more transformation information than projected embeddings $\mathbf{z}$. The relative performance of $\mathbf{h}$ and $\mathbf{z}$ may depend on the downstream objective.
- **Batch Size and Temperature.** We expect performance to degrade as batch size decreases due to fewer in-batch negatives. Consistent with alignment–uniformity theory [9], higher temperatures may partially mitigate this degradation

Experiment	Stage	Configs	Seeds	Epochs	Hours/Run	Total GPU Hours
Baseline (SimCLR)	Confirm	1	3	800	4	12
Augmentation Ablation	Screen	4	1	200	1	4
Projection Head Variants	Screen	3	1	200	1	3
Batch $\times$ Temperature Sweep	Screen	12	1	100	0.5	6
MoCo Comparison	Screen	1	1	200	1	1
Top 2 Findings Re-run (CIFAR-10)	Confirm	2	3	800	4	24
Optional: STL-10 Confirmation	Confirm	2	1	800	4	8
Total (Core Plan)	–	–	–	–	–	**50**
Total (With STL-10)	–	–	–	–	–	**58**

Table 1: Experimental plan: screening runs use reduced epochs and one seed; confirmatory runs use full 800-epoch training with three seeds. GPU-hour estimates assume ResNet-18 on an NVIDIA A100.

by smoothing the contrastive objective. Screening sweeps should reveal systematic trends across (N, τ) combinations.

- **Momentum-Based Alternative.** We expect MoCo to exhibit reduced sensitivity to batch size relative to standard SimCLR, supporting the interpretation that large batches are primarily a mechanism for increasing negative sample diversity rather than a fundamental requirement.

Across experiments, alignment and uniformity metrics will help interpret observed accuracy differences through changes in embedding geometry.

Overall, our findings aim to distinguish between design choices that are inherently necessary and those whose importance depends on dataset scale and computational regime.

6 Risks and Limitations

While our staged experimental protocol improves computational efficiency, it introduces several potential risks.

Reduced-Epoch Screening Bias. Screening experiments use 100–200 epochs, which may not fully capture long-term convergence behavior. Some configurations that appear suboptimal in early training could perform better under full 800-epoch schedules. To mitigate this risk, we re-train selected high-signal configurations in the confirmatory stage.

Limited Dataset Scale. Our experiments focus on CIFAR-10 and STL-10, which are significantly smaller than ImageNet. While this is intentional to study low-resource regimes, conclusions may not fully transfer to large-scale settings.

Batch–Temperature Grid Resolution. To control compute cost, we use a reduced grid over batch size and temperature. Finer-grained sweeps might reveal more precise optima. Our goal, however, is to identify systematic trends rather than exact hyperparameter maxima.

Representation Evaluation Protocol. We rely primarily on linear evaluation and alignment–uniformity metrics. While widely used, these metrics may not capture all aspects of representation quality, such as transfer to structured prediction tasks.

Implementation Sensitivity. Contrastive learning methods are sensitive to implementation details (augmentation strength, normalization, optimizer tuning). Although we follow established settings, small deviations may affect reproducibility across hardware environments.

7 Future Work

Over the next two weeks, we will focus on validating key findings from the screening stage through full confirmatory training.

Baseline Confirmation. We will complete full 800-epoch baseline training on CIFAR-10 across three random seeds and report mean and standard deviation of linear evaluation accuracy to establish a stable reference point.

Augmentation Confirmatory Study. We will re-train the most informative augmentation configurations (e.g., full pipeline vs. crop-only) using full 800-epoch schedules and three seeds to determine whether screening-stage trends persist under full convergence.

Projection Head Validation. We will conduct confirmatory training for the default two-layer projection head and the strongest alternative configuration, using full training and three seeds. We will also complete linear probing experiments to assess transformation information retained in $\mathbf{h}$ and $\mathbf{z}$.

8 Conclusion

We present a systematic re-evaluation of SimCLR under small-scale and compute-constrained settings. Using a staged experimental protocol, we isolate the effects of augmentation composition, projection head design, and batch size–temperature interaction.

Rather than proposing a new method, our goal is to assess the robustness of SimCLR's design principles outside large-scale ImageNet training. Our results aim to clarify which components remain essential in low-resource regimes and which are primarily artifacts of scale and compute.

By combining controlled ablations with geometric diagnostics, this study provides practical guidance for applying contrastive learning under realistic constraints.

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